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SUPERSEDING
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FEDERAL SPECIFICATION

INSULATION SLEEVING, THERMAL (URETHANE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rigid cellular urethane fabricated or premolded into standard shapes for pipe, tube, and vessel insulation.

1.2 Classification.

1.2.1 *Types.* Expanded urethane thermal insulation shall be of the following types, as specified (see 6.2).

Type I—Preformed Self-extinguishing.
Type II—Preformed Nonburning.

1.2.2 *Forms, sizes, and thicknesses.*

1.2.2.1 *Forms.* Covering for standard forms shall be straight elbows, tees, or unions. Special forms (vessels) shall be curved segments or beveled lags, cut to fit curved surfaces, as specified (see 6.2). Pipe forms are split in half lengthwise for sizes up to and including 8 inches. Forms for sizes larger than 8 inches may be in sections.

1.2.2.2 *Size and thickness.* Covering shall be of the following sizes and thicknesses, as specified (see 6.2):

Tube O.D.—From 3/8 to 6-1/8 inches.
Covering thickness—From 1/2 to 2-1/2 inches.
Nominal pipe size—From 1/4 to 12 inches.
Covering thickness—From 1/2 to 4 inches.

1.2.3 *Length.* Unless otherwise specified (see 6.2), covering shall be furnished in 36 inch lengths.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-640—Boxes, Fiberboard, Corrugated, Triple-Wall.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Hand-

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books and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

American Society For Testing And Materials (ASTM) Publications:

C 177—Test for Thermal Conductivity of Materials by Means of the Guarded Hot Plate.

C 302—Test for Density of Preformed Pipe-Covering-Type Thermal Insulation.

C 335—Test for Thermal Conductivity of Pipe Insulation.

C 355—Test for Water Vapor Transmission of Materials Used in Building Construction.

D 756—Test for Resistance of Plastics to Accelerated Service Conditions.

D1621—Test for Compressive Strength of Rigid Cellular Plastics.

D 1692—Test for Flammability of Plastic Foams and Sheeting.

D 2127—Test for Water Absorption of Rigid Cellular Plastics.

E 171—Specification for Standard Atmospheres for Conditioning and Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103.)

3. REQUIREMENTS

3.1 Material. Thermal insulation shall be made basically from the reaction of an organic polyisocyanate with a polyol resin, using a suitable expanding agent, and shall be rigid in composition and essentially a closed cell structure.

3.2 Tolerances, both types. When tested in accordance with 4.4.2.1, insulation shall conform to the following tolerances:

Length—Plus or minus 1/32 inch per linear foot.

Thickness—Plus or minus 1/16 inch.

Inside dimensions—Plus 1/16 inch on standard outside pipe dimensions up to 8 inch size. Plus 1/8 inch on standard outside pipe dimensions for 8 inch and larger size.

3.3 Thermal conductivity. The thermal conductivity (k) for all types of insulation measured at a mean temperature of 75°F. shall not exceed 0.016 Btu in/hr. sq. ft. °F., or at a mean temperature of 40°F. shall not exceed 0.15 Btu in/hr. sq. ft. °F. (see 4.4.3).

3.4 Dimensional stability, both types.

3.4.1 *Low temperature.* When tested in accordance with 4.4.4.1, the change in length between bench marks on test specimens shall not exceed minus 2 percent. Any sample tested showing evidence of cell collapse or visible distortion in any axis shall be rejected.

3.4.2 *High temperature.* When tested in accordance with 4.4.2, the change in length between bench marks on test specimens shall be not more than plus 4 percent. Any sample tested showing evidence of cell collapse or visible distortion in any axis shall be rejected.

3.4.3 *Humid aging.* When tested in accordance with 4.4.4.3, the change in length between bench marks on test specimens shall be not more than plus or minus 5 percent. Any sample tested showing evidence of cell collapse or visible distortion in any axis shall be rejected.

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3.5 Flammability. Type I material shall achieve a rating of self-extinguishing and type II material a rating of nonburning (see 4.4.5).

3.6 Compressive strength, both types. The compressive strength in the direction of thickness of the pieces shall be not less than 20 p.s.i. at the maximum compressive load or at 10 percent deflection, whichever occurs first (see 4.4.6).

3.7 Water absorption, types I and II. Water absorption shall be not more than 2 percent by volume (see 4.4.7).

3.8 Moisture vapor permeability, types I and II. Moisture vapor permeability shall be not more than 2.0 perm inches (see 4.4.8).

3.9 Density. Density of both types shall be not greater than 2.5 (PCF) (see 4.4.9).

3.10 Workmanship. Insulation shall be free of voids, and accumulation of unexpanded material or other occlusions; free from dirt on any surface; and free from objectionable odor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 *Lot.* Unless otherwise specified (see 6.2), a lot shall consist of all insulation pieces of the same type and size offered for delivery at the same time.

4.2.2 *Sampling for dimensional and visual examination of end item.* Sampling of insulation for these characteristics shall be in accordance with level S-3.

4.2.3 *Sampling for tests.* Samples for test shall be in accordance with level S-2.

4.3 Examination.

4.3.1 *End item.* Insulation shall be examined for defects listed in table I. The acceptable quality levels (AQL's) in accordance with MIL-STD-105 shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.1.1 *Thickness and length of insulation.* The thickness and length of insulation shall be measured using a rule graduated in 1/32 inch increments.

4.3.1.2 *Inside dimensions.* Measurements of clearances shall be made by means of feeler gauges designed to suitably measure clearances between curved surfaces, and read to the nearest 1/16 inch.

TABLE I. *Classification of defects, end item*

Defects	Major	Minor
Type not as specified	X	
Holes extending through thickness of insulation exceeding 1/4 by 1/4 inch per 3 lineal feet	X	
Breakage not exceeding 50 sq. inches		X
Crushing of not more than 10 percent of original thickness on more than 10 percent of actual size of individual piece	X	
Fissures	X	
Length, thickness, and inside dimensions not within tolerances	X	
Torn, crushed, or ragged end or edges	X	
Mated edges of section out of alignment by more than 1/16 inch		X
Not free from grease, oil, dirt, or other foreign matter		X

4.3.2 *Inspection of preparation for delivery.* An inspection shall be made to determine that packaging, packing, and mark-

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ing requirements of section 5 are complied with. Defects shall be scored in accordance with table II. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE II. *Classification of preparation for delivery defects*

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate or improper packaging or packing, such as, contents not close fitting in interior container; container flaps or taping of seams, corners, and manufacturers' joint; interior containers not close fitting; loose strapping, not as specified; inadequate stapling; bulging or distortion of containers.
Container	Number per container not as specified.
Weight (exterior container)	Weight per container exceeds maximum specified.

4.4 **Tests.**

4.4.1 *Acceptance.* The AQL for tests shall be 4.0 percent defective.

4.4.2 *Specimens.*

4.4.2.1 *Conditioning before tests.* Specimens shall be conditioned for 24 hours at 74° ± 3°F. (23° ± 1.7°C.) and 50 ± 2 percent relative humidity in accordance with ASTM E 171 after 14 days from date of manufacture.

4.4.2.2 *Thickness.* One test specimen 3 inches wide by 8 inches long selected from each sample unit shall be measured in at least three places with a dial gage micrometer, or equivalent, and the thickness taken as the average of three readings. The micrometer shall have a flat presser foot with an area of 1 square inch. The other flat bearing surface shall not be smaller than the presser foot. The presser foot shall exert a force of four ounces. The gage shall be graduated in 0.001 inch increments (see 3.2). If the average thickness reading of the test specimen is not within the tolerances specified in 3.2, the sample unit shall be rejected.

4.4.2.3 *Linear dimensions.* A scale graduated to 0.01 inch shall be used for measuring the length of test specimens. The length shall be read from the scale to the nearest 0.03 inch. Each linear dimension of the test unit shall be measured at three equally spaced places (see 3.2).

4.4.2.4 *Inside dimensions.* Each sample unit shall be measured for inside dimensions (see 4.3.1.2). If the readings of inside dimensions are not within tolerances specified in 3.2, the sample unit shall be rejected.

4.4.3 *Thermal conductivity.* The thermal conductivity of insulation materials devoid of any covering shall be determined by the guarded hotplate method described in ASTM C177, at a mean temperature of 75° or according to ASTM C 335, after the test specimens have been conditioned for 12 days at 90° ± 3°F. relative humidity (see 3.3).

4.4.4 *Dimensional stability, both types.*

4.4.4.1 *Low temperature.* Test shall be made in accordance with ASTM D 756, procedure V, conducting low temperature test only (see 3.4.1).

4.4.4.2 *High temperature.* Test shall be made in accordance with ASTM D 756, procedure II, conducting high temperature test only. During tests, volume change shall be recorded by measuring samples every 24 hours. Duration of test shall be continued until maximum peak volume change is recorded. The dimensional stability shall be

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reported as change in length between bench marks, expressed as a percentage of the length measured initially (see 3.4.2).

4.4.4.3 *Humid aging.* Tests for humid aging shall be conducted in accordance with procedure I of ASTM D 756. Test results shall be recorded each 24 hours until a maximum peak of volume change is recorded (see 3.4.3).

4.4.5 *Flammability, types I and II.* Flammability of types I and II insulation shall be determined by ASTM D 1692 (see 3.5).

4.4.6 *Compressive strength.* The compressive strength of insulation shall be determined in accordance with method A in ASTM D 1621 (see 3.6).

4.4.7 *Water absorption.* The water absorption of insulation shall be determined in accordance with ASTM D 2127 (see 3.7).

4.4.8 *Moisture vapor permeability.* The moisture vapor permeability of insulation shall be determined by the desiccant method described in ASTM C 355 at 75° ± 3°F. and 50 ± 2 percent relative humidity.

4.4.9 *Density.* Density of both types shall be determined in accordance with ASTM C 302 (see 3.9).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Insulation of like description in quantities as specified (see 6.2) shall be packaged in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed in accordance with the appendix to the box specification. The gross weight shall not exceed the weight limitations of the box specification. The boxes shall have waterproof protection by use of tape.

5.1.2 *Level B.* Insulation of like description in quantities as specified (see 6.2) shall be packaged in a close-fitting box conforming to PPP-B-636, type CF, class domestic. The box shall be closed in accordance with the appendix to the box specification. The gross weight shall not exceed the weight limitations of the box specification.

5.1.3 *Level C.* The insulation shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Unit packages of insulation of like description shall be packed in close-fitting boxes conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight shall not exceed 150 pounds.

5.2.2 *Level B.* Unit packages of insulation of like description shall be packed in close-fitting boxes conforming to PPP-B-640, class 2, grade B. The boxes shall be closed and strapped in accordance with the appendix to the box specification. The gross weight shall not exceed 150 pounds.

5.2.3 *Level C.* The insulation shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Markings.

5.3.1 *Civil agencies.* In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* The addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 *Type I.* Type I insulation is intended for use where material is protected with

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noncombustible covering or facing in a temperature range of minus 30° to plus 200°F.

6.1.2 *Type II.* Type II insulation is intended for use where material may be exposed in use or where a higher degree of protection than that afforded by type I is desired for use in a temperature range of minus 30° to plus 200°F.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type (see 1.2.1).
(*c*) Form (see 1.2.2.1).
(*d*) Size and thickness (see 1.2.2.2).
(*e*) Length (see 1.2.3).
(*f*) Size of lot, if different from 4.2.1.
(*g*) Selection of the applicable levels of packaging and packing required (see 5.1 and 5.2).
(*h*) Quantity required in unit package (see 5.1.1).

6.3 Temperature limits. For specific applications, depending on humidity conditions and allowable linear expansion, the actual temperature limits should be agreed upon between the manufacturer and the purchaser.

6.4 Flammability. The types of foam covered by this specification are based on an ASTM test method, used as a small scale laboratory screening test for classifying flammability. The test is used solely to establish relative burning characteristics, and the classification should not be considered a fire hazard classification.

MILITARY CUSTODIANS:
Army—MO
Navy—YD
Air Force—69

Review activities:
Army—None
Navy—YD
Air Force—69

User activities:
Army—CE
Navy—AS, SS
Air Force—None

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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